

## A bullet hole recognition method based on visual detection

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**Abstract**—Shooting Intelligent Detection System has important applications in the military industry. Existing target shooting systems mainly face problems such as high cost, complex operation, and poor environmental adaptability. Considering the above limitations, in this paper we propose a fast, low-cost, and easy-to-deploy bullet hole recognition algorithm based on visual detection, which mainly consists of four parts: image preprocessing, extraction of target center, bullet hole extraction and ring value determination. In the process of bullet hole extraction, a method for segmenting effective target surface is proposed to cope with the problem of off-target bullet holes. Our method can not only accurately localize the bullet holes, but also obtain the exact ring value at that location. In addition, due to the robust adaptability of our method, it can even be applied to outdoor scenarios.

**Keywords**—image processing; bullet hole detection; shooting intelligent detection system; visual detection

### I. INTRODUCTION

Shooting practice is an important part of military training. Feedback on shooting results relies on manual target reporting, which is time-consuming, unsafe, and subject to human error. With the constant advancement of technology, It has become inevitable to design automatic bullet hole recognition systems with the ability to report automatic shooting results.

Automated target reporting systems<sup>[1]</sup> are more accurate, efficient and secure than manual target reporting. Hitherto, there are several types of automated target reporting systems based on their implementation principles: double-layer electrode short-circuit sampling system, photoelectric automatic target reporting system, acoustic-electric positioning automatic target reporting

system, fibre-optic coding automatic target reporting system and electrode-embedded automatic target reporting system. However, these systems are harshly configured and costly to maintain. With the continuous development of computer vision technology, many scholars have integrated visual detection into automatic target reporting systems. Generally speaking the automatic target reporting system based on visual detection is mainly composed of four parts: image pre-processing, target ring extraction, bullet hole recognition, and ring value determination, among which bullet hole recognition is the core of the whole automatic target reporting which directly determines the precision of the final target reporting. Many bullet hole recognition methods based on different approaches have been proposed. The template matching based bullet hole recognition method<sup>[2][3]</sup> is based on calculating the feature vectors of the bullet holes and comparing the similarity with the feature vectors of each pixel point in the image to be detected, and finally locating the bullet holes by similarity analysis. However, this method requires a large number of accurate bullet hole templates and is also computationally intensive. Fuzzy theory based on approach<sup>[4]</sup> identifies bullet holes on the basis of the difference that exists between the gray value of the bullet hole and the background gray value, which is simple and fast but does not take into account the target surface distortion. The time series image based on subtraction technique<sup>[5][6]</sup> identifies the bullet holes by comparing the two frames before and after and considering the position where the difference is not zero as a bullet hole. It is simple but vulnerable to environmental influences. Image information fusion technology based on wavelet transform<sup>[7]</sup> identifies bullet holes and performs wavelet decomposition on two images respectively. The wavelet

transform coefficients of the image with new bullet points are mainly changed to highlight the edge and position of the new bullet points. This approach improves immunity at the cost of complexity and time-consumption. The method of recognizing bullet holes based on their gray scale features<sup>[8]</sup> is to use the difference in gray scale properties between the bullet hole region and the background, and statistically extract the bullet holes by segmenting the gray scale features of the bullet holes, which is a simple algorithm but is greatly affected by the illumination.

In summary, automatic target reporting methods based on image processing<sup>[2]-[10]</sup> are relatively mature, but there is a certain gap in the speed and accuracy of recognition and anti-interference ability to the environment, compared with acoustic and electric positioning and electronic targets. In this regard, in this paper we propose a novel bullet hole recognition algorithm based on visual detection, which can achieve fast and accurate recognition of bullet holes.

## II. DESIGN OF BULLET HOLE RECOGNITION ALGORITHM BASED ON VISUAL DETECTION

### A. The overall design of the algorithm

The bullet hole recognition algorithm based on visual detection is mainly divided into four parts, which are image preprocessing, extraction of target center, bullet hole extraction and ring value determination. Image preprocessing is the correction of the image captured by the camera, and at the same time reduces the amount of information in the image as well as enhances the image features, and is a prerequisite for the later detection of bullet holes. Extraction of target center is mainly to extract the center position of the ten rings and its radius, which is the basic condition to judge the ring value where the bullet hole is located. The four components are interrelated and their linkages are shown in Figure 1 below.

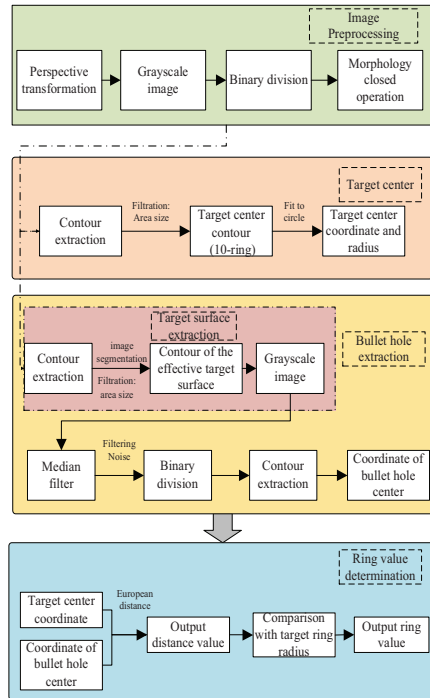


Figure 1. Overall design of the algorithm

### B. Image Correction

The image acquired from the camera requires some processing before it can be used for subsequent detection. The main purpose of image preprocessing is to eliminate extraneous information in the image, enhance the detectability of relevant information, and maximize the simplification of the data so as to improve the reliability of feature extraction and recognition. In order to avoid damage to the camera during target shooting, the camera is placed below the target paper and at an angle of elevation, so geometric correction of the image is required, as well as binary segmentation of the image and morphological operations. Image geometry correction uses perspective transformations. The perspective transformation is the process of projecting a two-dimensional image onto a three-dimensional viewing plane and then transforming it to two-dimensional coordinates as follows:

Let the original coordinate point be  $(x, y)$  and the transformed coordinate be  $(X, Y, Z)$ , then the perspective transformation matrix equation is expressed as:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = A \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (1)$$

Where  $A$  is the perspective transformation matrix, which can be split into four parts: The first part  $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$  is a

linear transformation and this part of the matrix is mainly used for scaling and rotating operations of the image; The second part  $[a_{31} \ a_{32}]$  is used for translation operations

and the third part  $[a_{13} \ a_{23}]^T$  is used to generate perspective transformations; The fourth part of the parameter is a fixed value of 1.

The coordinates  $(X, Y, Z)$  derived from the previous transformation matrix are three-dimensional coordinates, so they need to be divided by the Z-axis to transform to the two-dimensional plane to get the new two-dimensional coordinates  $(X', Y')$ , the results are as follows:

$$X' = \frac{a_{11}x + a_{12}y + a_{13}}{a_{31}x + a_{32}y + a_{33}} \quad (2)$$

$$Y' = \frac{a_{21}x + a_{22}y + a_{23}}{a_{31}x + a_{32}y + a_{33}} \quad (3)$$

Expanding the above equation with yields the following equation:

$$\begin{cases} a_{11}x + a_{12}y + a_{13} - a_{31}xX' - a_{32}yX' = X' \\ a_{21}x + a_{22}y + a_{23} - a_{31}xY' - a_{32}yY' = Y' \end{cases} \quad (4)$$

So solving the perspective transformation matrix requires 4 sets of point pairs. Perspective transformations are more flexible than linear and affine transformations, placing them in three-dimensional space enhances the dimensionality. Perspective transformations maintain

"straightness", which means that a straight line in the original drawing remains straight after a perspective transformation. The corrected image is shown below in Figure 3 and Figure 2 shows the perspective transformation matrix:

|          |         |         |
|----------|---------|---------|
| 0.4935   | 0.1217  | -466.88 |
| 0.01248  | 0.8122  | -316.83 |
| 8.84e-06 | 1.9e-04 | 1       |

Figure 2. Perspective transformation matrix



(a) Original image



(b) Corrected image

Figure 3. Image correction results

### C. Extraction of target center and Bullet hole extraction

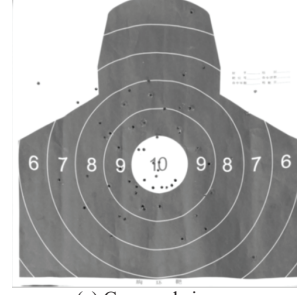
The original image was geometrically corrected to give an approximate image of the target area. To further obtain the target surface region of interest in the image, further preprocessing of the corrected image is required: gray processing, binary segmentation, and morphological operations.

#### 1) Image Preprocessing

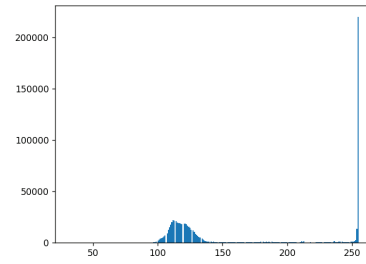
Image gray processing is the process of converting a color image into a grayscale image. The grayscale image contains only one channel, and the channel value represents the grayscale value. During gray processing, the RGB values of each pixel are combined into a single grayscale value. The three components were weighted and averaged with different weights based on importance and other indicators. Since the human eye is most sensitive to green and least sensitive to blue. Therefore, the weighted average of the three RGB components according to the following formula can get a more reasonable grayscale image. The gray scale image is shown in Figure 4 below:

$$Gray(i, j) = 0.299 * R(i, j) + 0.578 * G(i, j) + 0.114 * B(i, j) \quad (5)$$

Where  $R(i, j)$ 、 $G(i, j)$ 、 $B(i, j)$  represent the pixel values of the three color channels and  $Gray(i, j)$  represents the single channel pixel value after grayscale.



(a) Gray scale image



(b) Histogram of gray level

Figure 4. Result after gray processing

After the image is grayed out to be a 256 level gray image, it is then processed using binary segmentation<sup>[11][12]</sup> to obtain a binarized image that still reflects the image features by selecting the 256 brightness levels of the gray image through appropriate threshold values. The aggregate nature of the image relates only to the location of points with pixel values of 0 or 255, and no longer involves multiple levels of pixel values, making data processing simple. The process of binary partitioning is as follows:

Let the pixel value of a gray scale image be  $gray(x, y)$  and the threshold for segmentation be  $T$ .

The image  $d(x, y)$  after performing segmentation is defined as follows:

$$d(x, y) = \begin{cases} 0, & gray(x, y) < T \\ 1, & gray(x, y) \geq T \end{cases} \quad (6)$$

After the image is binarized, the binary image obtained will have a lot of noise and the boundary information of the target region is blurred, using morphological operations to process the binary image for image enhancement. The core operation of morphology is morphological operations, which are used to change the shape and characteristics of an image by means of specific operations of structuring element with the image. The structuring element is a small and predefined shaped image matrix that can perform logical operations with pixels in an image. Two common morphological operations are erosion and dilation. In gray scale morphology the image functions  $f(x, y)$  and  $b(x, y)$  are used to denote the target image and the structural elements, and

$(x, y)$  denotes the coordinates of the pixel points in the image. The mathematical expression is shown below, where  $g(x, y)$  is the erosion or dilation grayscale image:

Erosion:

$$g(x, y) = f \oplus b$$

$$= \max \{ f(x + x', y + y') - b(x', y') | (x', y') \in D_b \} \quad (7)$$

Dilation:

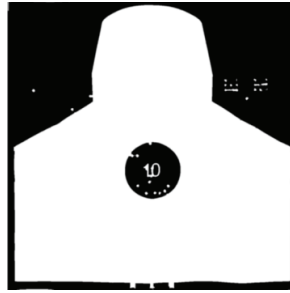
$$g(x, y) = f \otimes b$$

$$= \max \{ f(x + x', y + y') - b(x', y') | (x', y') \in D_b \} \quad (8)$$

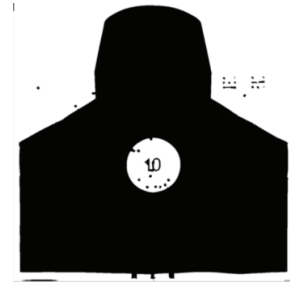
Erosion and dilation are usually paired together. Erosion followed by dilation is called the open operation, and is mainly used to remove some smaller targets, separate some finer joints, and smooth the boundaries of some larger objects; dilation followed by erosion is called the closure operation and is mainly used to fill small holes, connect nearby targets, and smooth out target boundaries. The inverse operation is performed on the binary map before the closure operation, and the value of the target surface area is set to "1" in order to highlight the target area to be processed. The missing parts of the target surface region in the binary map are then connected using closed operations to form a more complete target surface region, as shown in Figure 5 below.



(a) binarized image



(b) morphological image



(c) inverse image

Figure 5. Result after pre-processing

## 2) Target center and bullet hole detection

After the previous image correction and preprocessing of the original image, a binary image containing the target surface of the bullet hole is obtained. On this basis, the target paper is firstly contour extracted<sup>[13]</sup>, and the detected contours are eliminated according to the contour area in order to leave the middlemost ring. Then the detected contour is fitted to a circle to obtain the center of target coordinates and center of target radius, and the results are shown in Figure 6 below. The process of contour extraction and fitting is as follows:

Step one: Suppose the input image is  $F = \{f_{ij}\}$ . First raster scan is performed on the input binary image, from top to bottom and from left to right, to find points that satisfy the boundary conditions. If  $f_{i,j} = 1$  and  $f_{i,j-1} = 0$ , then  $(i, j)$  is the starting point of the outer boundary;

Step two: After finding the start point of the boundary trace, perform the boundary trace to find the complete boundary;

Step three: After obtaining the complete boundary, take the center point of the pixels of the boundary of the connected domain and connect them to become a contour. Finally use Green's formula to calculate the area of the contour:

$$\oint_L Pdx + Qdy = \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy \quad (9)$$

Step four: The obtained contour area is filtered to leave the target contour. Then the obtained irregular contour is fitted to a circle and the center coordinates and its radius are obtained.



Figure 6. Result of target center extraction

Prior to bullet hole detection, the valid target surface needs to be extracted to eliminate the effect of off-target bullet holes on detection. The previously acquired binary image is first contour extracted, and the effective target surface is screened by area to generate a mask. The original image will then be segmented according to the region of the mask to leave the final target region for bullet hole detection and the results are shown in Figure 7 below.



(a) Mask area



(b) Target area

Figure 7. Effective target surface extraction

A similar strategy is adopted for the detection of bullet holes, where the previously acquired effective target surface image is first gray scaled to obtain a grayscale image. Then the median filter is applied to the grayscale image in order to eliminate noises such as ring lines and background folds, and then a suitable threshold is selected for binarization. The contour detection of the binarized image does not circle the edges of the bullet holes in a complete and continuous manner, there will be a break in the edge line, which occurs when a morphological closure operation needs to be done. Finally, the detected bullet hole contours are fitted to the smallest circle, while the center of mass coordinates of each bullet hole are obtained. The results are shown in Figure 8 below.

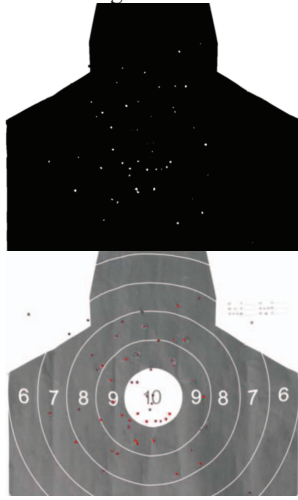


Figure 8. Results of bullet hole detection

#### D. Ring value judgement

In order to determine later the value of the ring where the bullet hole is located, the main idea of this algorithm is to use the principle of the concentric characteristics of the rings in the chest-ring target and the radius difference between the rings. After the previous processing, the target center, the radius of the target center and the coordinates of each bullet hole have been obtained. The Euclidean distance between each bullet hole and the center of the target is first calculated using the Euclidean distance, and then the distance is compared with the radius of the circle in which each ring value is located. For bullet holes on the target line are usually counted on the side with the highest

ring value, and the higher ring value will be the valid score. The judgement formula is as follows:

$$d_i(x, y) = \sqrt{(x_r - x_i)^2 + (y_r - y_i)^2} \quad (10)$$

$$S = \begin{cases} 10, & d_i \leq r \\ 9, & r < d_i \leq 2r \\ 8, & 2r < d_i \leq 3r \\ 7, & 3r < d_i \leq 4r \\ 6, & 4r < d_i \leq 5r \\ 5, & 5r < d_i \leq 6r \end{cases} \quad (11)$$

Where  $(x, y)$  represents the coordinates of the pixel points in the image,  $i$  represents the number of bullet holes, and  $S$  represents the ring value.

### III. RESULTS AND ANALYSES OF EXPERIMENTS

The accuracy of bullet hole detection directly determines the reliability of the automatic target reporting system. The accuracy of the bullet hole detection algorithm based on visual detection designed in this paper is mainly determined by two parts: one is the extraction of the effective target surface; the other is the accuracy of bullet hole recognition. The main factors affecting the accuracy of bullet hole recognition are the quality of the camera shot, the lighting and the density of the bullet holes. In order to better count the scores, this algorithm marks the bullet holes with colors to distinguish the ring values they are in. The relationship between the color and ring value settings is shown in Table 1.

The results of the 10 target face images were counted in the test, and the ring values of all the bullet holes in each target face were summed as the results for that target face. Some of the image results are shown in Figure 10, and the total results are shown in Figure 9. The numbers on the left side of Figure 9 represent the statistical scores, the numbers on the right side represent the accuracy, and the lowest side represents the serial number of the test target paper.

TABLE I. RELATIONSHIP BETWEEN BULLET HOLES AND COLOR

| Ring value | Color  |
|------------|--------|
| 10         | Red    |
| 9          | Yellow |
| 8          | Cyan   |
| 7          | Purple |
| 6          | Green  |
| 5          | Pink   |

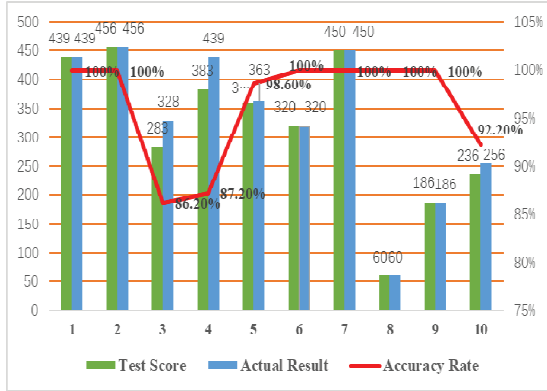


Figure 9. Statistics on achievements

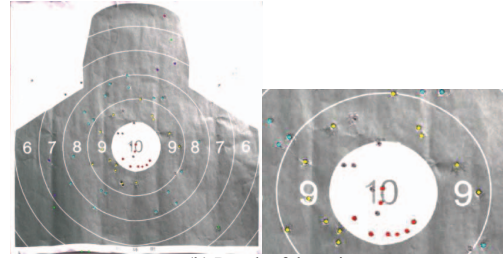


Figure 10. Result of bullet hole detection

In the bullet hole recognition algorithm designed in this paper, the difference between the grey value of the bullet hole and the region in which it is located, the greater the difference between the grey values, the higher the recognition accuracy. During the test, it was found that when the target paper is subjected to direct sunlight during the day, the target surface will be overexposed, which will cause a decrease in accuracy, as shown in Figure 11 below. Previously recognisable bullet holes in the middle part of the target paper were not detected due to the effect of light, which had a significant impact on the recognition results. For this problem, this algorithm incorporates a filter with bands in  $532nm \pm 10nm$  in front of the image acquisition device to eliminate the effect of visible light in the rest of the bands, which improves the accuracy of the recognition. The results after adding the filter are shown in Figure 12 and 13 below. The numbers on the left side of Figure 13 represent the statistical scores, the numbers on the right side represent the accuracy, and the lowest side represents the serial number of the test target paper.



(a) Original image



(b) Result of detection

Figure 11. Exposure phenomena



(a) Filtered image



(b) Result of detection

Figure 12. Result after filtering

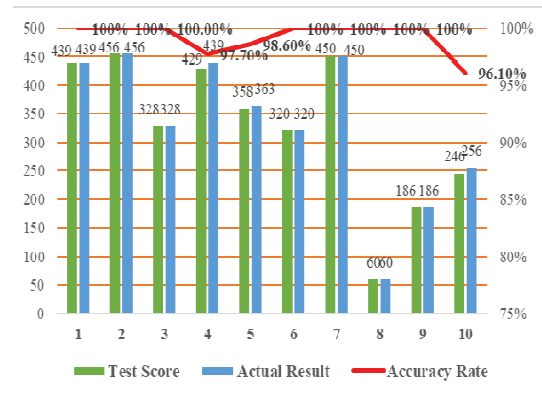


Figure 13. Statistics on achievements

#### IV. CONCLUSION

The automatic target reporting system will be the mainstream way to replace manual target reporting during shooting training in the future. In this paper, a bullet hole recognition algorithm based on visual detection is proposed, which consists of four main parts: image

preprocessing, extraction of target center, bullet hole extraction and ring value determination.

In addition, in order to reduce the effect of outdoor light on recognition, this method adds a filter in front of the image acquisition device to filter out visible light of a certain wavelength. The experimental results show that the algorithm can effectively detect the bullet holes in the target ring and calculate the scores within the allowable range of error, and can be used in outdoor environments. In future work, we consider improving the accuracy of detecting dense bullet hole situations, as well as extracting effective target surfaces more quickly.

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